



ASSIGNMENT 2

CUSTOMER & MARKETING ANALYTICS

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Executive Summary

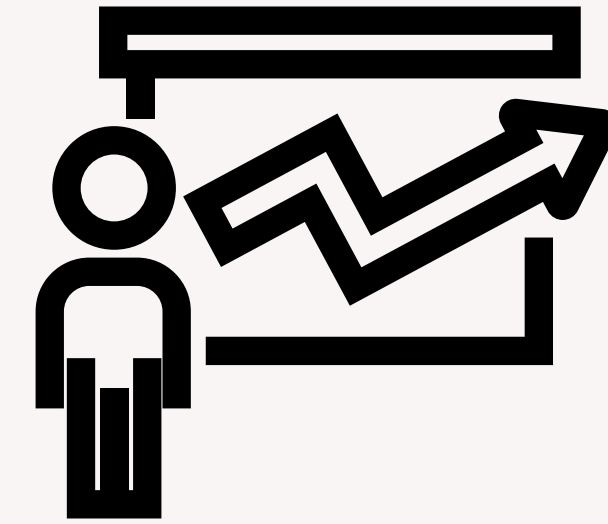


General Structure

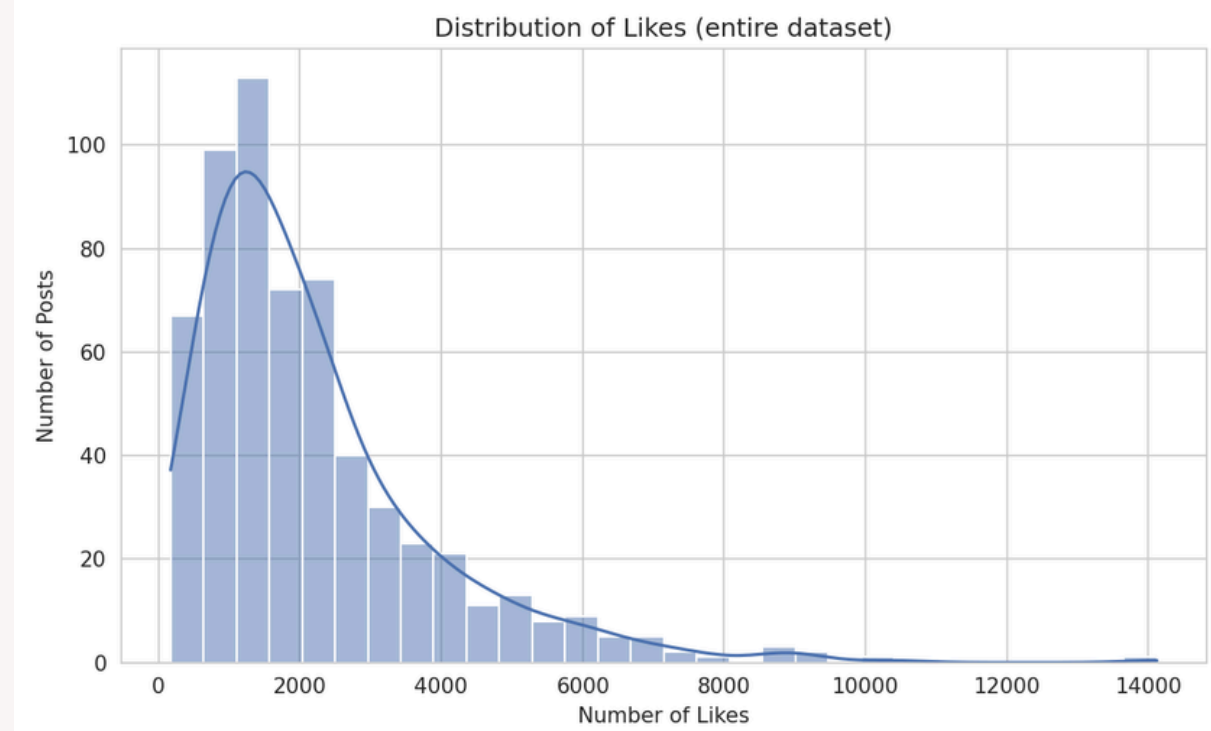
- 600 posts split evenly between LumoraBeauty, FitPulse and TechNest
- 145 unique posts: 51 by LumoraBeauty, 46 by FitPulse and 48 by TechNest
- 9 starting features
- No missing values

index	Post_ID	Brand	Handle	Post_Text	Total_Followers	Total_Posts_LastYear	Num_Likes	Num_Comments	Num_Reposts
356	357	FitPulse	@FitPulseOfficial	Our monthly newsletter includes a new training plan and nutrition tip every issue. Subscribe via link in bio.	312000	276	1778	55	82
484	485	TechNest	@TechNestHQ	Tech people have two moods: this is the most elegant solution I have ever seen, and who wrote this code 🤖💻 Drop your latest reaction below 🗨️⚡	218000	194	1117	58	461
298	299	FitPulse	@FitPulseOfficial	What does your ideal rest day look like? Walk us through it 🗨️🧘 Recovery is part of the training!	312000	276	1849	60	544
473	474	TechNest	@TechNestHQ	Cheap USB-C cables can damage your devices 🚫 Poorly manufactured cables may deliver incorrect voltage. Always buy cables that meet USB-IF certification standards ⚠️	218000	194	493	15	42
230	231	FitPulse	@FitPulseOfficial	RPE (Rate of Perceived Exertion) is a 1 to 10 scale to measure workout intensity without a heart rate monitor. Aim for RPE 7 to 8 for most strength training sets.	312000	276	2623	33	224

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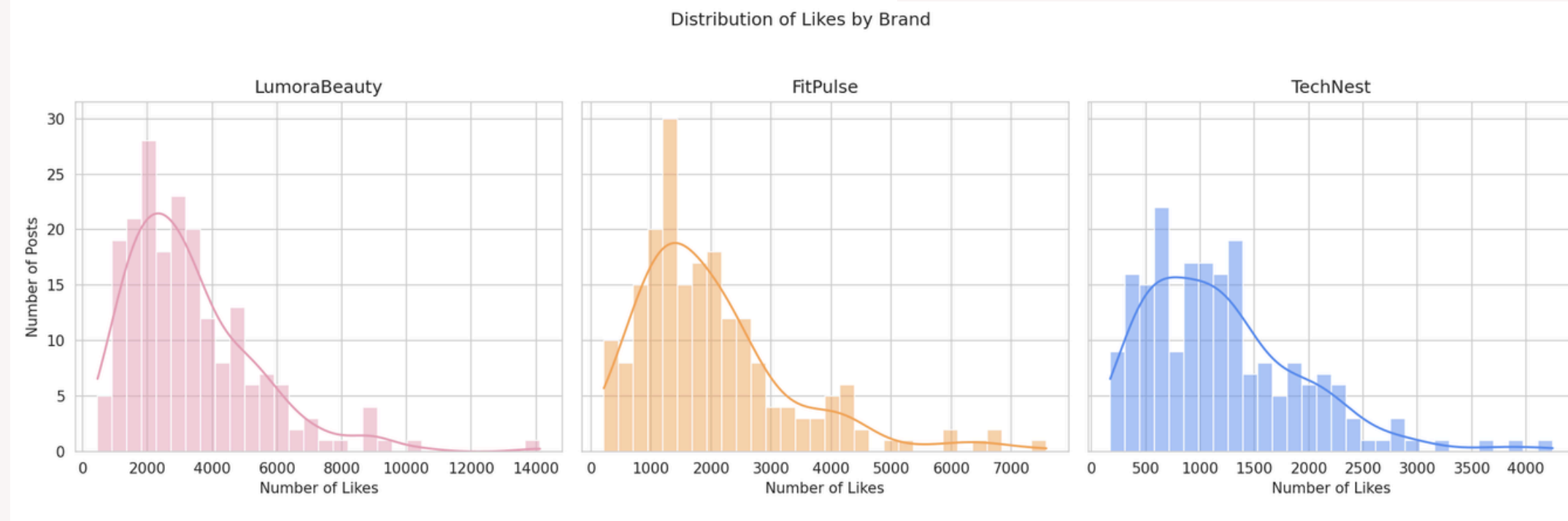


Engagement Metric: Likes

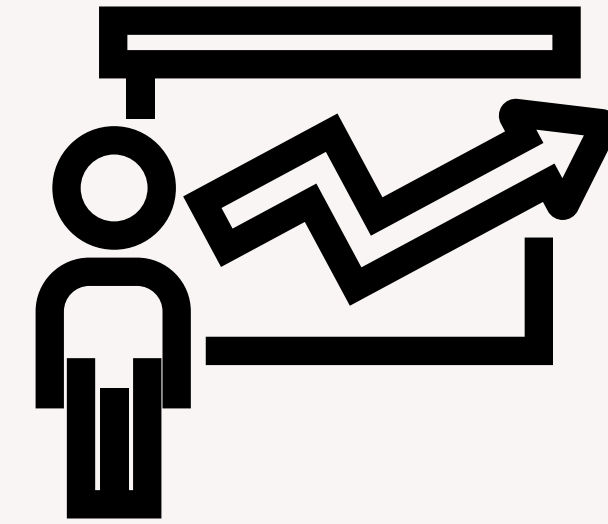


Stats

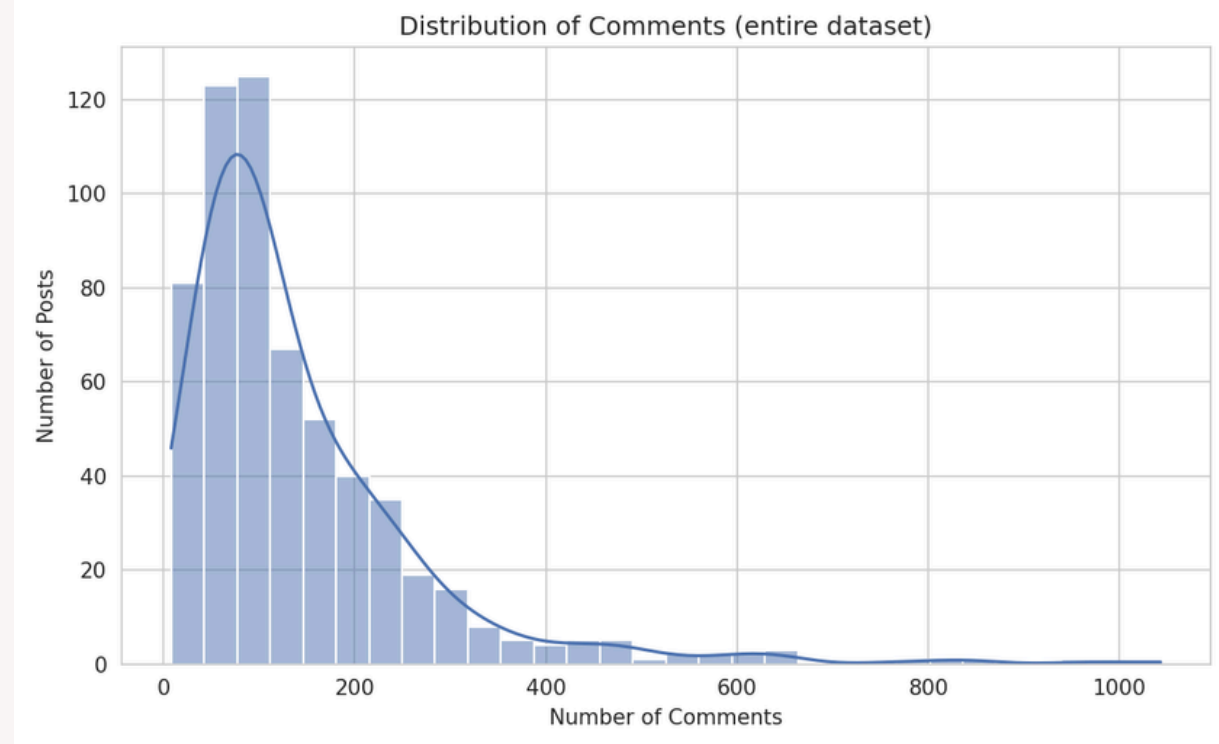
- Mean: 2186.7
- std: 2186.7
- min: 173
- max: 14121



Executive Summary

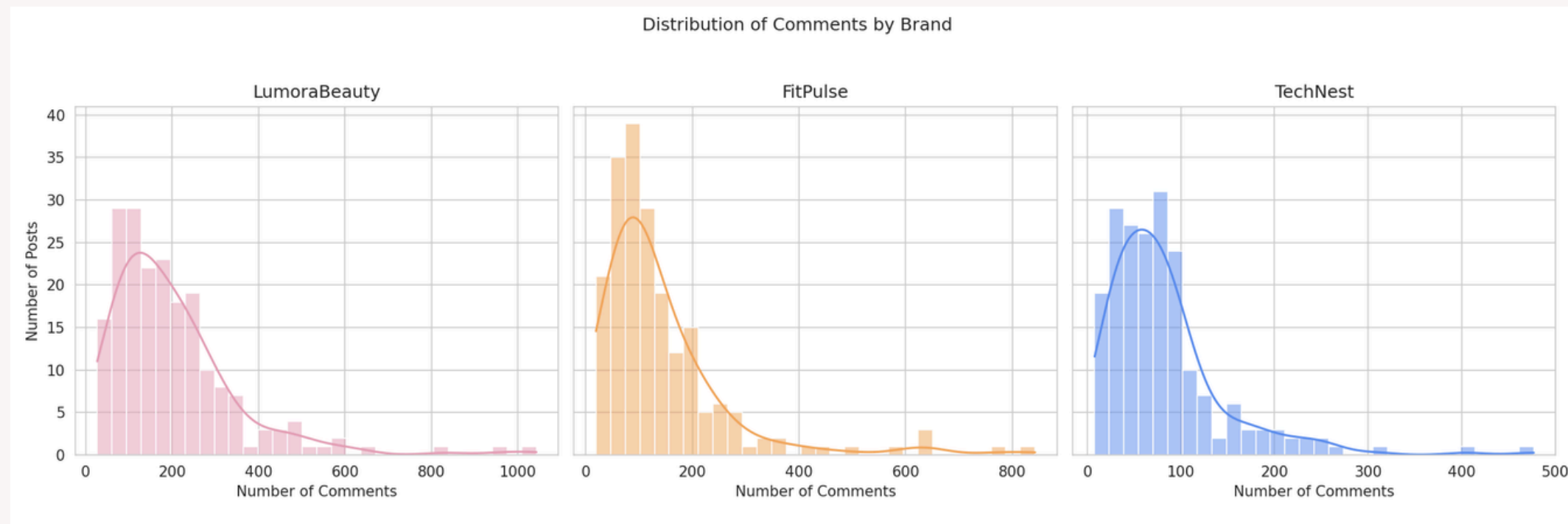


Engagement Metric: Comments

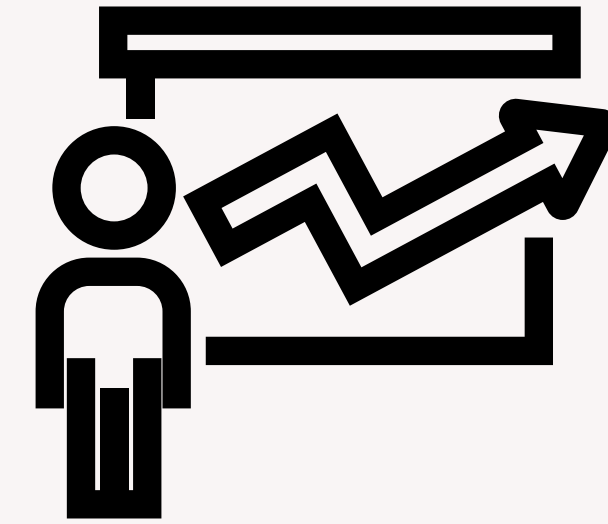


Stats

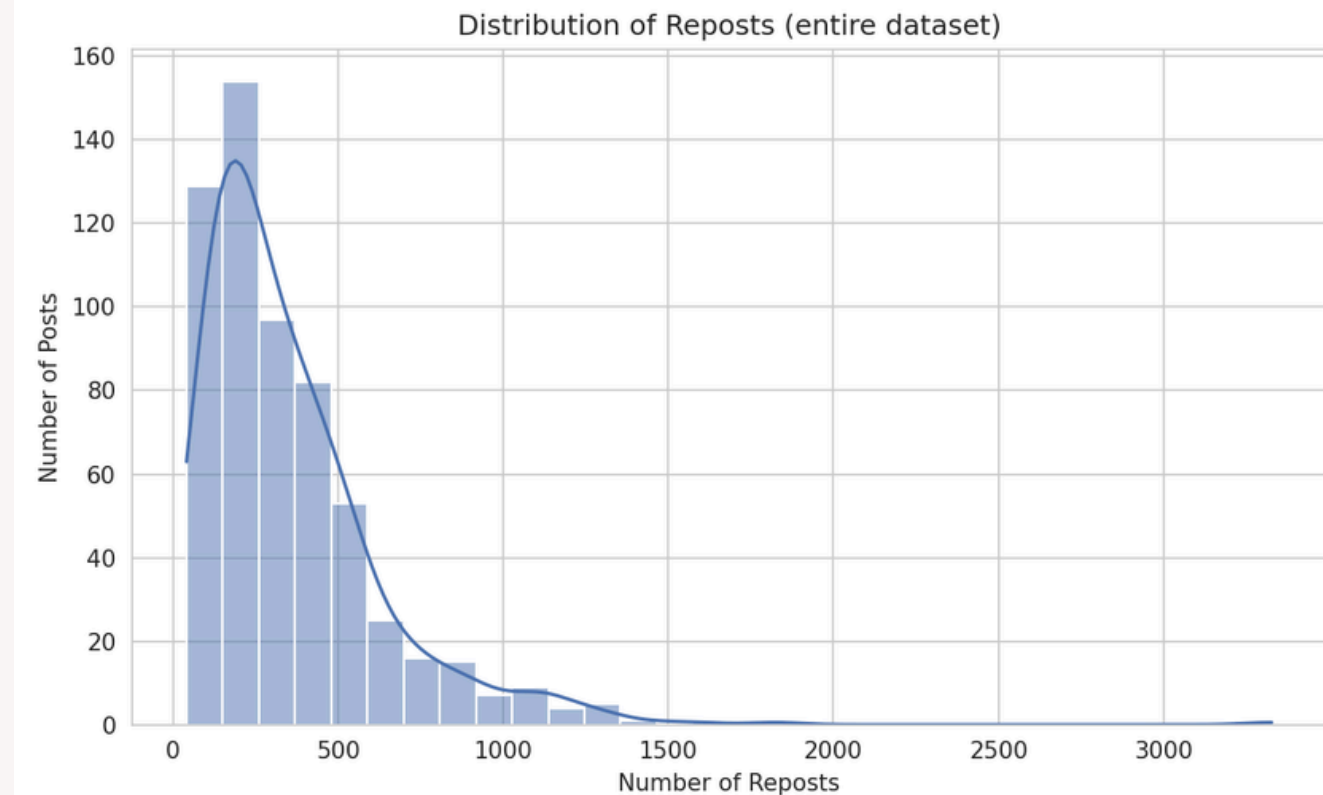
- Mean: 142.2
- std: 129.8
- min: 102
- max: 1043



Executive Summary

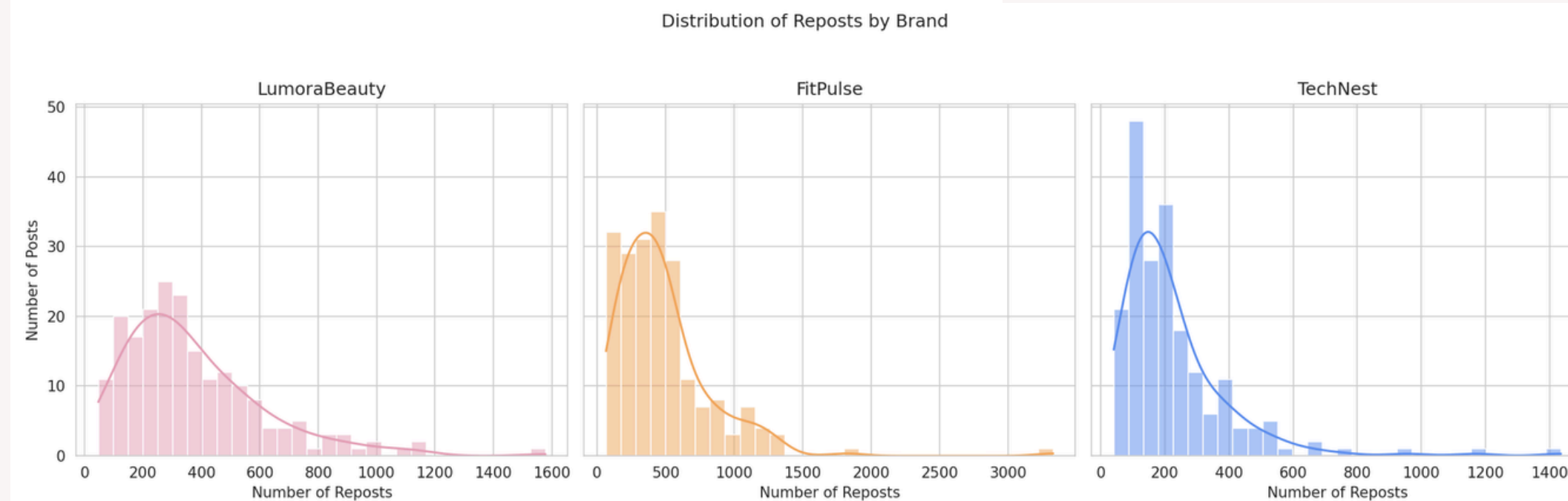


Engagement Metric: Reposts



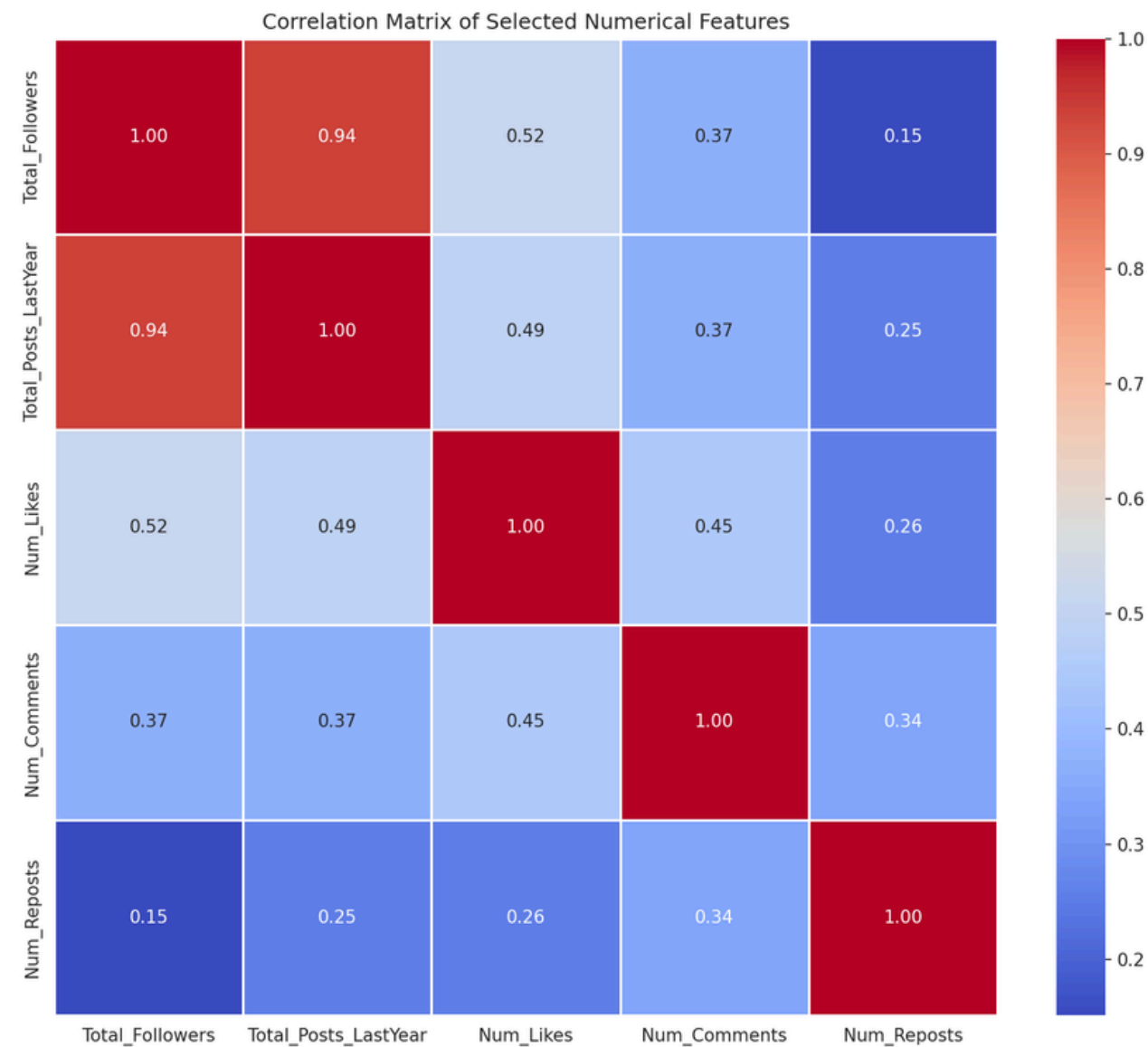
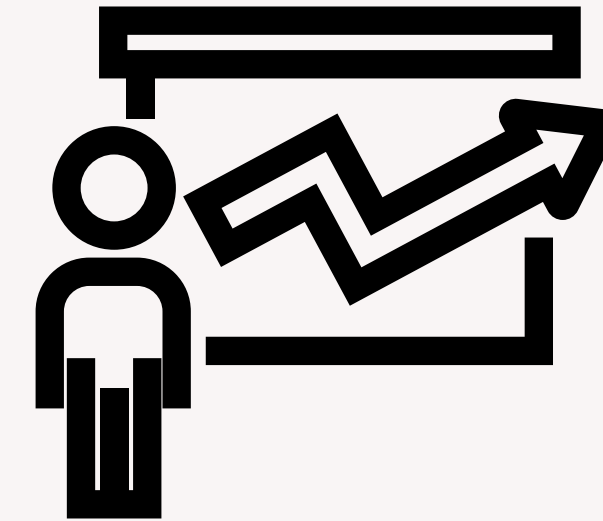
Stats

- Mean: 359.4
- std: 291.9
- min: 41
- max: 3327



Executive Summary

Feature Correlation



- **Followers vs. Activity (0.94):** There is a near-perfect positive correlation between Total_Followers and Total_Post_LastYear. This is highly plausible, as accounts with a large following are typically established profiles that maintain high levels of publishing activity.
- **Inter-Engagement Correlation (0.45):** Num_Likes and Num_Comments show a moderate positive correlation. This indicates that content which resonates visually or emotionally (generating Likes) also tends to spark conversation and discussion among the audience.
- **Impact of Audience Size:** The Total_Followers feature has a significant impact on engagement, correlating at 0.52 with Num_Likes and 0.49 with Num_Comments. Simply put, a larger reach increases the probability of generating interactions.
- **Impact of Posting Consistency:** A similar, though more moderate, effect is observed for Total_Post_LastYear, which correlates at 0.37 with both Likes and Comments. This suggests that a consistent posting strategy helps maintain brand visibility and keeps the audience involved over time.

Question 1

Sentiment Analysis

+0.19

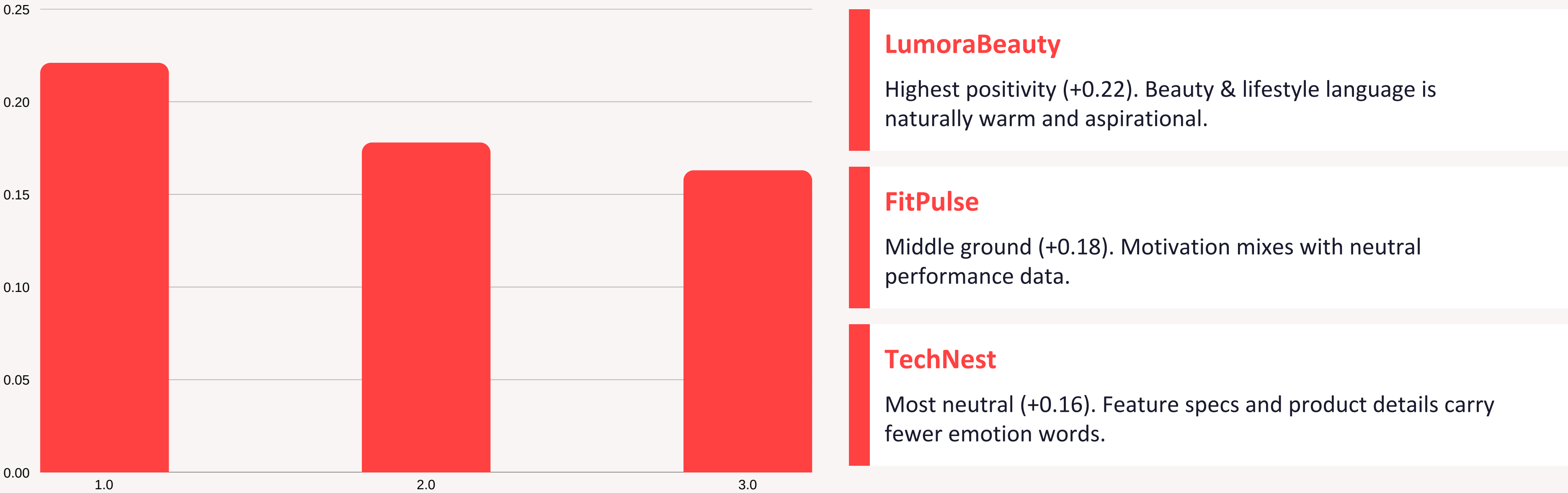
Overall mean polarity

~30%

Posts score exactly 0.0

All +

Every brand is positive



Key takeaway: All brands are positively toned. The ~30% zero-score rate reflects neutral product posts — not negative sentiment.

Question 2

Topic Classification

3 Labels

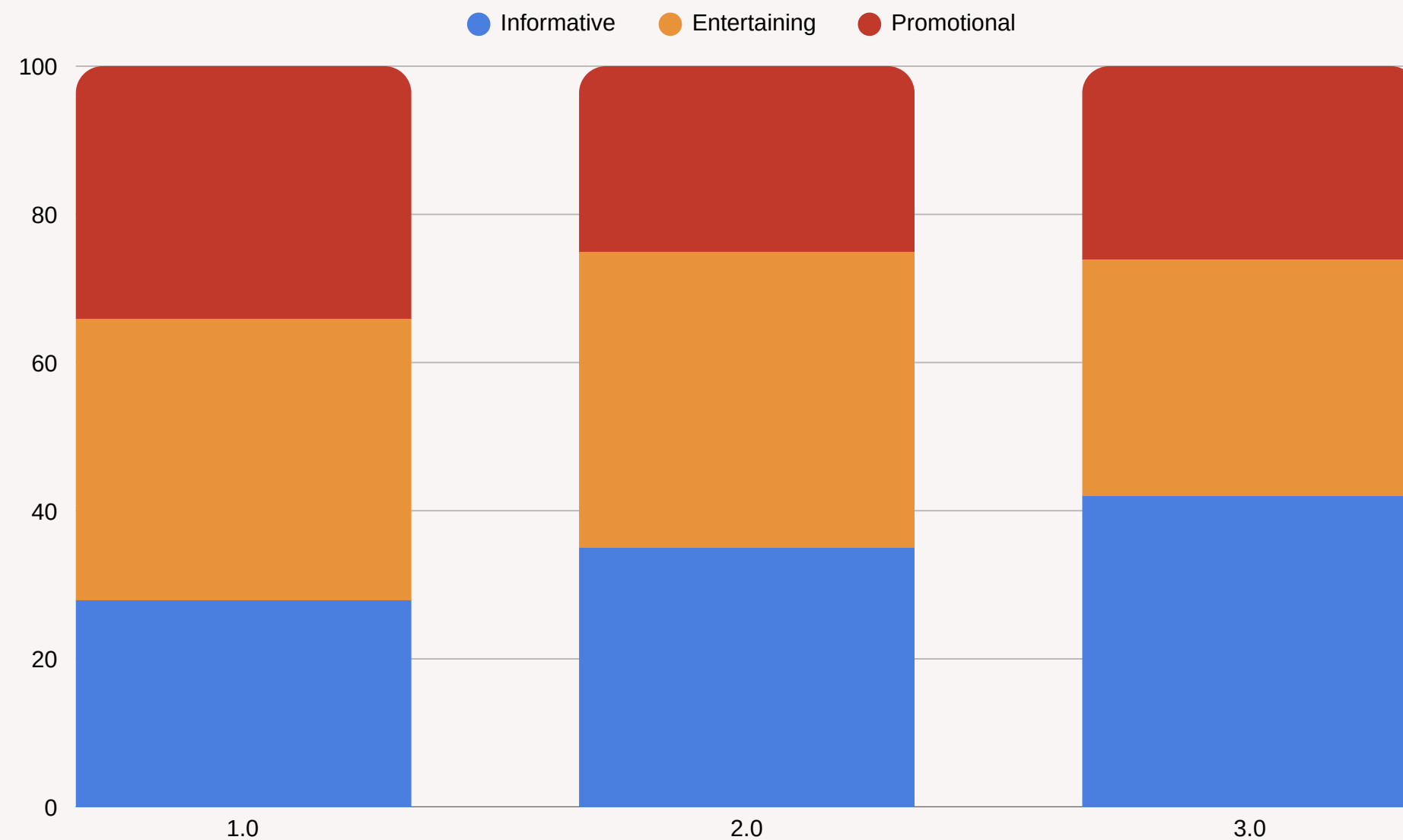
Informative · Entertaining · Promotional

V1 → V3

3 iterations to improve accuracy

BART

Zero-shot — no training data needed



LumoraBeauty

34% Promotional — highest of all brands. Heavy on discounts, launches & CTAs.

FitPulse

Most balanced mix. Entertaining content leads at 40% across all 3 topics.

TechNest

42% Informative — highest of all brands. Specs, how-tos & feature guides dominate.

Key takeaway: Each brand has a distinct content strategy. V3 (BART + keyword rules) gives the most accurate classification.

Question 3

Emoji analysis : counts, types, brand comparison

Emoji Count

- Overall average: 2.4 emojis/post — max never exceeds 5
- Most posts fall between 2 and 4 emojis
- Correlation with sentiment: +0.018 → purely a stylistic choice

Count by Brand

- TechNest: avg. 2.56 (median 3) — heaviest user
- LumoraBeauty: avg. 2.44 (median 3)
- FitPulse: avg. 2.30 (median 2) — most restrained

Top Emojis by Brand

- LumoraBeauty: ✨ sparkles, ❤️ yellow_heart, 🌸 cherry_blossom → aspirational/beauty
- FitPulse: 💪 flexed_biceps, 🔥 fire, 📵 pointing_down → energetic/fitness
- TechNest: 💻 laptop, ⚡ high_voltage, 📵 pointing_down → informative/tech
- Only shared emoji across all brands: 📵 (call-to-action)

Question 4

Drivers of Engagement — What Makes Posts Perform?

Model Choice

Negative Binomial Regression

Why? Likes & comments are count data.

Variance/mean ratio:

- Likes: 1,328×
- Comments: 119×

Far exceeds Poisson threshold → NB is the right model.

Predictors Used

- Sentiment_Score
- Topic_Class
- Emoji_Count
- Has_Face_Emoji
- Text_Length
- Brand dummies

Model Fit

Metric	Likes	Comments
Pseudo R ²	0.036	0.045
AIC	9,938.5	6,759.0
Observations	600	600

KEY FINDINGS (Incidence Rate Ratio — IRR > 1 boosts, IRR < 1 hurts)

0.64×

FitPulse vs Lumora
(likes)

36% fewer likes

0.36×

TechNest vs Lumora
(likes)

64% fewer likes

1.08×

Entertaining
vs Informative

8% more likes

0.97×

Face Emoji Present

3% fewer likes (n.s.)

+0.02

Emoji Count
per extra emoji

~2% lift (n.s.)

Key Insight: Brand identity (follower base) dominates content variables. Community size drives engagement far more than what you post or how you say it.

Marketing Implications for BrandPulse

1. Grow the Community First

Brand dummies dwarf NLP variables. A post from LumoraBeauty (485K followers) generates ~2.8× the likes of an identical TechNest post (218K followers). Follower growth is the highest-leverage investment.

2. Entertaining Content Outperforms Informative

Entertaining posts get ~8% more likes than informative ones across all brands. Pure product-information posts are the weakest content type - always blend in engagement hooks.

3. Emoji Strategy Matters More Than Quantity

Raw emoji count shows a small positive lift but not statistically significant at dataset level. However face emojis (😂😄) slightly hurt performance - brands should favour object/action emojis over expressive faces.

4. Sentiment Alone Doesn't Move the Needle

Sentiment score is not a significant predictor for any brand individually. Being more positive doesn't reliably generate more likes. Focus on content type and emoji choice instead.

Question 5

Brand Differences - Do Engagement Drivers Differ Across Brands?

Method 1: Interaction model (Brand × NLP terms) - LR test $p = 0.172 \rightarrow$ NOT significant | Method 2: Separate NB model per brand — compare IRRs

PER-BRAND IRR COMPARISON (Num_Likes outcome, vs Informative baseline)

Predictor	LumoraBeauty	FitPulse	TechNest	Significant?
Sentiment Score	0.870	1.108 ↑	0.829	None ($p > 0.20$)
Emoji Count	1.263 ↑	1.076	0.985	Lumora only ($p=0.01$)
Entertaining topic	0.875	1.150	1.110	None ($p > 0.10$)
Promotional topic	0.851	1.173 ↑	1.122	None ($p > 0.10$)
Face Emoji	0.811 ↓	0.983	1.050	Lumora hurts ($p=0.07$)

LumoraBeauty

Emoji count is the only significant NLP driver (IRR = 1.26 per emoji). Face emojis hurt. Keep sparkles ✨💛, avoid 😂😭.

FitPulse

Sentiment points positive (IRR=1.11) & promotional content shows a small lift — but neither is significant. Directionally, positive energy works.

TechNest

No NLP variable is significant. Engagement driven almost entirely by the follower gap vs Lumora. Focus on growing the community.

Verdict: Drivers are broadly consistent across brands (LR $p=0.172$). A unified content strategy works - but Lumora should prioritise emojis, FitPulse lean positive, TechNest invest in community growth.

Question 6

Managerial implications and recommendations

Key Engagement Drivers

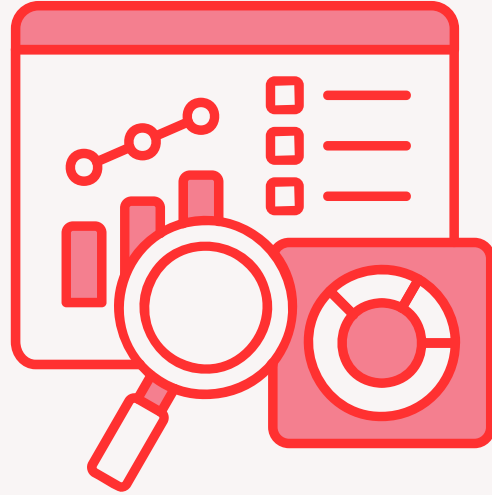
- +1 emoji → +21% likes, +26% comments (strongest driver across all brands)
- Community size matters more than content: FitPulse & TechNest get up to 64% fewer likes than LumoraBeauty
- Sentiment and topic category have no meaningful impact on engagement

Brand Recommendations

- **LumoraBeauty**: keep emojis high, shorten captions (–4% likes per extra word), avoid smiley faces
- **FitPulse**: don't force positivity — focus on relevant, effort-related emojis over quantity
- **TechNest**: long captions are fine — prioritise community posts and maintain high emoji count

Is More Always Better?

- More emojis help, but type matters as much as quantity
- Wrong emoji type can penalise engagement (e.g. face emojis hurt LumoraBeauty)



Appendix

- Extensive doc: <https://docs.google.com/document/d/1kPT-1PPfIZ-bXVOa6yqEU6SdHHrlkhElzX3f0rvTYXw/edit?usp=sharing>
- Code: https://drive.google.com/file/d/1vEHBwv_ppilh14sCWOYJIDcBatkE9hPP/view?usp=drive_link

**Thank you for
your attention !**